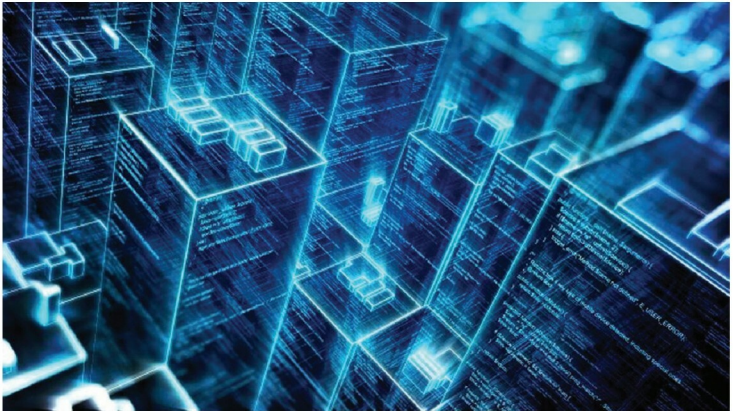


A Primer on Software Defined Networking (SDN) and the OpenFlow Standard

Continuous innovation and the need to adapt to the constraints of conventional networking has made software defined networking (SDN) pretty popular. It is an approach that disassembles the network's control plane and data plane. This allows network administrators to program directly without having to worry about the hardware specifications.



OpenFlow, the first SDN standard, is a communication protocol in software defined networking (SDN). It is managed by the Open Networking Foundation (ONF). The SDN controller or the 'brain' interacts with the forwarding (data) plane of the networking devices like routers and switches via OpenFlow APIs. It empowers the network controllers to decide the path of network packets over a network of switches. The OpenFlow protocol is required to move network control out of exclusive network switches and into control programming that is open source and privately overseen.

Software-defined networking uses southbound APIs and northbound APIs. The former are used to hand over information to the switches and routers. OpenFlow is the first southbound API. Applications use the northbound APIs to interact.

Porting an OpenFlow switch in ns-3

The OpenFlow 1.3 module for ns-3, widely known as the OFSwitch13 module, was intended to boost the ns-3 network

simulator with SDN technology. An OpenFlow switch is a package that routes packets in the SDN environment. The data plane is referred to as the switch and the control plane is referred to as the controller. The OpenFlow switch interacts with the controller and the switch is managed by the controller via the OpenFlow protocol.

The fundamental components of the OpenFlow switch (as shown in Figure 2) incorporate at least one flow table, a meter table, a group table and an OpenFlow channel to an exterior controller. The flow tables and group table perform the packet scanning and forwarding function based on the flow entries configured by the controller. The routing decisions made by the controller are deployed in the switch's flow table. The meter table is used for the measurement and control of the rate of packets.

Configuring the SDN OFSwitch

In this article, we have incorporated the OFSwitch13 (version 1.3) with ns-3. To benefit from the features of OFSwitch13, an